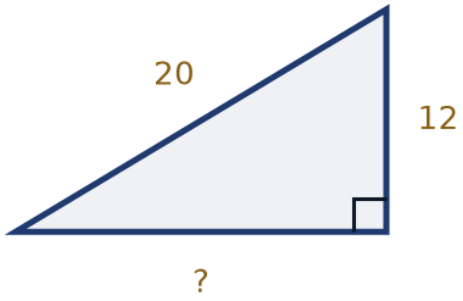
Trigonometry & Additional Topics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

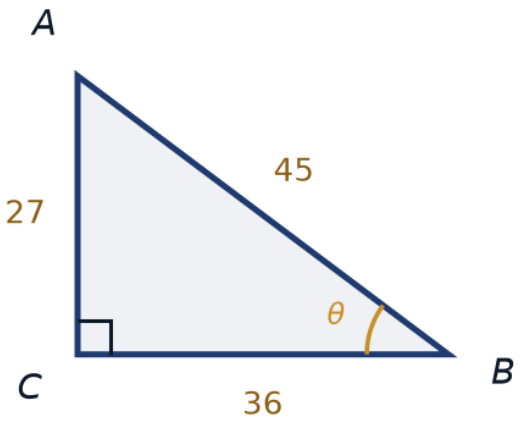
1 [Foundational] • Target: Right-triangle trigonometry

In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.



2 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.



3 [Foundational] • Target: Right-triangle trigonometry

A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

4 [Foundational] • Target: The Pythagorean identity

Given $\sin\theta = \frac{5}{13}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

5 [Foundational] • Target: The Pythagorean identity

Given $\cos\theta = \frac{8}{17}$ with θ acute, find $\sin\theta$.

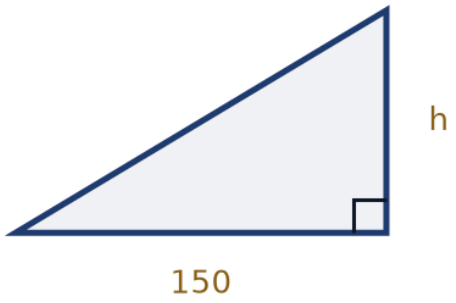
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6 [Foundational] • Target: Exact trigonometric values

State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.

7 [Foundational] • Target: Solving for missing sides and angles

A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32°, as shown below. Find the height of the tower, to the nearest foot.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving for missing sides and angles

From the top of a 60-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

9 [Intermediate] • Target: Complex numbers: arithmetic

Simplify i^7, using $i^2 = -1$.

10 [Intermediate] • Target: Complex numbers: modulus and conjugates

Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

11 [Intermediate] • Target: Vectors: magnitude and operations

Vector $v = (3, 4)$. Find its magnitude.

12 [Intermediate] • Target: Vectors: magnitude and operations

Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.

13 [Intermediate] • Target: Vectors: magnitude and operations

Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

14 [Intermediate] • Target: Radian measure

Convert 150° to radians, in terms of π.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Radian measure** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

Convert $\frac{7\pi}{4}$ radians to degrees.

16 **[Challenging]** • **Target: A well-designed word problem: measuring the width of a river** • **Cross-Lesson: 1.01 Heart of Algebra**

This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C. (a) At point C, the angle between the bank and the line of sight to the tree is 35°. Use the tangent ratio to find AB, the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24°. Use this second measurement to find a second estimate of AB, to the nearest meter, and state which measurement gives the larger estimate.

17 **[Challenging]** • **Target: Radian measure** • **Cross-Lesson: 2.01 Quadratics & Polynomials**

In a right triangle, acute angle θ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

18 **[Challenging]** • **Target: A well-designed word problem: measuring the width of a river** • **Cross-Lesson: 3.01 Functions & Graphs**

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.

19 **[Challenging]** • **Target: Right-triangle trigonometry** • **Cross-Lesson: 4.01 Ratios, Rates & Data**

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

20 **[Challenging]** • **Target: The Pythagorean identity** • **Cross-Lesson: 5.01 Statistics & Probability**

If $\cos(2x + 10^\circ) = \sin(3x - 15^\circ)$, find the value of x and evaluate $\tan(x + 16^\circ)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Right-triangle trigonometry**

In a right triangle, the hypotenuse is 87 and the side opposite angle θ is 60. Find $\sin\theta$.

a. $\frac{21}{29}$

b. $\frac{20}{29}$

c. $\frac{20}{21}$

d. $\frac{62}{87}$

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22 [Foundational] • Target: The Pythagorean identity

In a right triangle, the side opposite angle θ is 6 and the side adjacent is 8. Find $\tan\theta$.

a. 1

b. $\frac{4}{5}$

c. $\frac{3}{5}$

d. $\frac{3}{4}$

23 [Foundational] • Target: Exact trigonometric values

In a right triangle, the hypotenuse is 25 and the side adjacent to angle θ is 24. Find $\cos\theta$.

a. $\frac{26}{25}$

b. $\frac{24}{25}$

c. $\frac{7}{24}$

d. $\frac{7}{25}$

24 [Foundational] • Target: Solving for missing sides and angles

In a right triangle, angle A and angle B are acute angles such that $\sin(A) = \frac{7}{11}$. What is the value of $\cos(B)$?

a. $\frac{7}{11}$

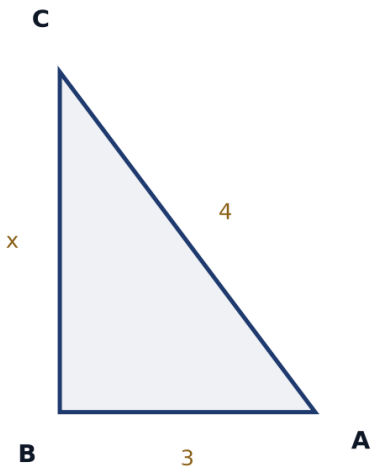
b. $\frac{4}{11}$

c. $\frac{8}{11}$

d. $\frac{11}{7}$

25 [Foundational] • Target: Complex numbers: arithmetic

In the right-angled triangle ABC shown below, $\angle ABC = 90^\circ$, $AB = 3$, and $BC = 4$. What is the length of the hypotenuse AC?



a. 5

b. 7

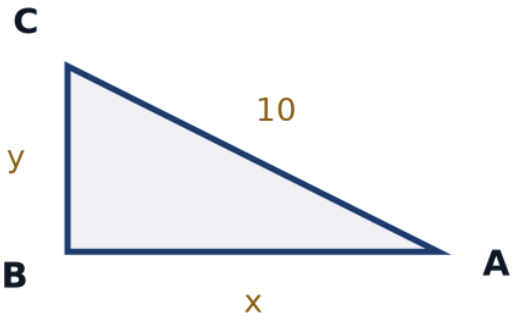
c. 12

d. 25

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26 [Foundational] • Target: Complex numbers: modulus and conjugates

In the right-angled triangle shown, what is the length of side AC if $AB = 10$ and $\angle B = 30^\circ$?



a. 5

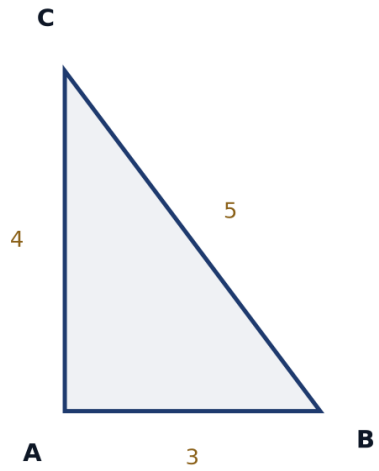
b. 10

c. $5\sqrt{3}$

d. 20

27 [Foundational] • Target: Vectors: magnitude and operations

In the right-angled triangle shown below, what is the value of $\sin(\theta)$?



a. $\frac{4}{5}$

b. $\frac{3}{5}$

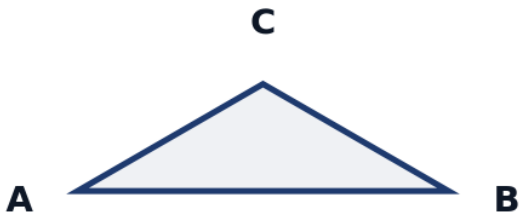
c. $\frac{5}{3}$

d. $\frac{5}{4}$

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28 [Foundational] • Target: Radian measure

In a right triangle, the hypotenuse is 10 and one angle is 30°. What is the length of the side opposite the 30° angle?



a. 6

b. 5

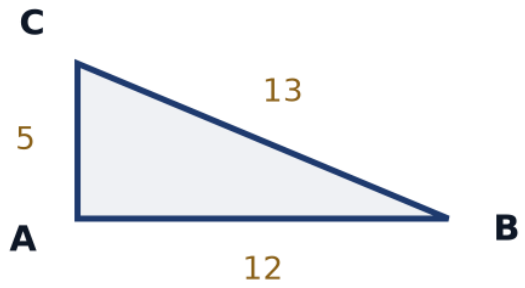
c. 10

d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: measuring the width of a river

In a right triangle, $\sin\theta = \frac{5}{13}$. What is $\cos\theta$?



a. $\frac{5}{12}$

b. $\frac{12}{13}$

c. $\frac{13}{12}$

d. $\frac{5}{13}$

30 [Intermediate] • Target: Right-triangle trigonometry

What is the radian measure of an angle of 330°?

a. $\frac{11\pi}{6}$

b. $\frac{\pi}{33}$

c. $\frac{330\pi}{360}$

d. $\frac{330}{180\pi}$

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31 [Intermediate] • Target: The Pythagorean identity

Given that $i = \sqrt{-1}$, what is the product of the complex numbers $(2 + 4i)(3 - 1i)$?

a. $2 + 10i$

b. $6 - 4i$

c. $10 + 14i$

d. $10 + 10i$

32 [Intermediate] • Target: Exact trigonometric values

In a right triangle ABC where angle $C = 90^\circ$, if $\cos(A) = \frac{12}{13}$, what is the value of $\tan(A)$?

a. $\frac{5}{12}$

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{13}{12}$

33 [Intermediate] • Target: Solving for missing sides and angles

In a right triangle, the side opposite angle θ is 18, the adjacent side is 24, and the hypotenuse is 30. What is the sine of θ?

a. $\frac{3}{4}$

b. $\frac{5}{3}$

c. $\frac{4}{5}$

d. $\frac{3}{5}$

34 [Intermediate] • Target: Complex numbers: arithmetic

A 30-60-90 right triangle has a short leg of length 4. What is the length of the long leg?

a. 8

b. $4\sqrt{2}$

c. $4\sqrt{3}$

d. $5\sqrt{3}$

35 [Intermediate] • Target: Complex numbers: modulus and conjugates

A 45-45-90 right triangle has legs of length 7. What is the length of the hypotenuse?

a. $8\sqrt{2}$

b. $7\sqrt{2}$

c. 14

d. $7\sqrt{3}$

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36 [Intermediate] • Target: Vectors: magnitude and operations

The diagram shows a triangle ABC, where D is a point on side AB and E is a point on side AC. It is given that DE is parallel to BC. If AD = 6 cm, DB = 4 cm, AE = 9 cm, and DE = 12 cm, find the length of the side BC.

a. 18 cm

b. 20 cm

c. 24 cm

d. 30 cm

37 [Intermediate] • Target: Radian measure

What is the dot product of the vectors $\vec{u} = \langle 0, 6 \rangle$ and $\vec{v} = \langle -5, -5 \rangle$?

a. 30

b. -30

c. -4

d. -25

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: measuring the width of a river • Cross-Lesson: 3.01 Functions & Graphs

If $\tan(\theta) = -\frac{3}{4}$ and $\sin(\theta) > 0$, what is the exact value of $\cos(\theta) - \sin(\theta)$?

a. $-\frac{7}{5}$

b. $-\frac{1}{5}$

c. $\frac{1}{5}$

d. $\frac{7}{5}$

39 [Challenging] • Target: Right-triangle trigonometry • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the magnitude of the vector $\vec{v} = \langle 9, 40 \rangle$?

a. 42

b. 49

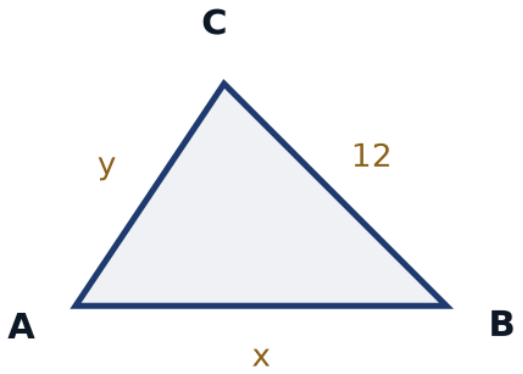
c. 41

d. 40

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40 [Challenging] • Target: The Pythagorean identity • Cross-Lesson: 5.01 Statistics & Probability

In the diagram below, ΔABC is inscribed in a circle. AB is a diameter. The length of chord AC is 12 units and the angle $\angle ABC = \theta$. If $\sin(\theta) = \frac{3}{5}$, find the length of the diameter AB.



a. 15 units

b. 20 units

c. 25 units

d. 30 units

41 [Challenging] • Target: Exact trigonometric values • Cross-Lesson: 6.01 Geometry

The terminal side of an angle θ in standard position passes through the point $(-3, 4)$. What is $\sin(\theta)$?

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $-\frac{4}{3}$

d. $\frac{3}{5}$

42 [Challenging] • Target: Solving for missing sides and angles • Cross-Lesson: 8.01 Exponentials & Logarithms

If $\sin(\theta) = \frac{7}{25}$ and θ is acute, what is $\cos(\theta)$?

a. $\frac{25}{25}$

b. $\frac{25}{24}$

c. $\frac{7}{25}$

d. $\frac{24}{25}$

43 [Challenging] • Target: Complex numbers: arithmetic • Cross-Lesson: 1.01 Heart of Algebra

Convert an angle of 45° into radians in simplest terms.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{\pi}{2}$

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44 [Challenging] • Target: Complex numbers: modulus and conjugates • Cross-Lesson: 2.01 Quadratics & Polynomials

A right triangle has legs of length $5\sqrt{3}$ and 5. What is the measure of the smallest angle in degrees?

a. 30°

b. 45°

c. 60°

d. 15°

45 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 3.01 Functions & Graphs

The three interior angles of a triangle measure 45°, 65°, and x°. What is the value of x?

a. 70

b. 80

c. 60

d. 75

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